

ProductionManagementAI

Today the application lets an automobile-parts plant create production orders (製造指示), find them, and watch progress on a dashboard. This page maps what it could do next, and in what order it makes sense to build it.

Where the application is today

SCR-001 製造指示 登録・編集 Create and edit an order; status path 下書き → 進行中 → 完了 / 取消; field locking and stale-save protection	SCR-002 製造指示一覧 Filter, sort and page every order; shareable view in the URL; overdue marker	SCR-003 ダッシュボード Status and delivery figures, overdue and due-soon lists, workload and completion charts, health check
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What it deliberately leaves out so far — each is a starting point below:

- Products change only through the database seed; there is no product screen
- No record of who changed an order, or when
- Two placeholder roles; everyone signed in can do everything
- An order records *that* it was completed, not how many good parts were made
- No production lines, plant calendar or capacity
- Nothing leaves the app: no export, print, notification or integration

Potential functions

Rough size, before any design: S days M 1–2 weeks L several weeks

Master data マスタ管理

The reference data every other function needs.

Product master 製品マスタ A screen to add, rename and retire parts (SKU, name, unit, drawing number), instead of changing the seed. Retired parts stay on old orders but leave the pick list. Builds on The existing products table and 製品 select on SCR-001 Needs Who may edit masters (role matrix)	Production lines 生産ライン・工程 Lines or work centers (e.g. 2号ライン) with their capacity per day, and which parts each line can make. Orders get a line. Builds on Product master Unlocks Load chart, schedule, dashboard by line
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Planning 生産計画

From "what is due" to "what can we make, where, and when".

Line load chart

負荷山積み

M

Open quantity per line per week against its capacity, highlighting overloaded weeks — the next step from today's workload-by-due-week chart.

Builds on

Dashboard workload chart

Needs

Production lines, plant calendar

Schedule board

生産スケジュール（ガント）

L

Orders laid out on a timeline per line; drag to reschedule, with warnings when a move would miss the due date or overload the line.

Needs

Production lines, plant calendar, start dates on orders

Material requirements

所要量計算

L

From open orders and the BOM, what materials are needed by when, compared with stock — a first, simple MRP.

Needs

Bill of materials, stock figures

Lot traceability

ロットトレーサビリティ

L

Record which material lots went into which order and which parts were shipped, so a defect can be traced both ways — expected by automotive customers (IATF 16949).

Needs

Bill of materials, results entry

Visibility and output 見える化・出力

Getting information to people, on screen and off.

Dashboard drill-down

ドリルダウン

S

Click an overdue order, a status tile or a chart bar to open the matching filtered list on SCR-002. The dashboard was deliberately kept read-only (WI-004 DEC-006); this would revisit that choice.

Builds on

SCR-002's filters already live in the URL

Dashboard by period and line

期間・ライン別表示

M

Choose the period (week, month, custom) and narrow by product or line; compare with the previous period.

Builds on

The single-snapshot dashboard query

Export

CSV・Excel 出力

S

Download the current list view — with its filters and sort — as CSV or Excel for reports and meetings.

Builds on

SCR-002's query endpoint

Printed work order

製造指示書の印刷

S

A one-page PDF per order with part, quantity, due date, notes and a QR code, to travel with the parts on the line.

Unlocks

Shop-floor scanning

Alerts

M

通知

A daily digest and immediate alerts for orders that become overdue or are due soon, by email or Teams, to the people responsible.

Builds on The dashboard's overdue and due-soon rules

Needs A deployed server; who is responsible for an order

Access and integration 権限・連携

Making it safe and connected enough for real use.

Role and permission matrix

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権限設計

Decide who can create, edit, cancel, complete and administer — e.g. operators record results but cannot cancel — and enforce it on every screen and API. An open project decision today (WI-001 DEC-015).

Builds on Server-side role checks already on every endpoint

Unlocks Masters, results entry, approval

Approval before release

M

承認フロー

A planner submits a Draft order, a supervisor approves it before production starts; rejected orders go back with a comment.

Needs Role matrix, change history

Company sign-in

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シングルサインオン

Sign in with the company account (e.g. Microsoft Entra ID) instead of app-specific passwords, with roles taken from company groups.

Builds on ASP.NET Core Identity cookie sign-in

ERP and customer orders

L

ERP・受注連携（EDI・内示）

Receive customers' forecast and firm orders and create production orders from them; send completions back to the ERP.

Needs Deployment host; a chosen ERP / EDI format

Multiple plants and English UI

M

複数工場・英語表示

Plants with their own time zone and calendar, and an English interface next to Japanese — all UI text is already in one catalog, so a second language needs no screen changes (WI-005 DEC-001).

Builds on The UI text catalog; the plant-clock abstraction

Production deployment

M

本番環境

Move from the local Docker Compose setup to a hosted environment with backups, monitoring and an image registry. An open project decision (WI-001 DEC-012); a prerequisite for alerts and integrations.

Builds on Existing container images, CI and OpenTelemetry

A sensible order to build them

Grouped so that each wave only depends on what came before. The order is a suggestion to discuss, not a plan.

Wave 1

Foundations

- Role and permission matrix
- Product master
- Change history
- Production deployment

Settles two open decisions and makes the data trustworthy.

Wave 2

Execution

- Production results entry
- Printed work order
- Export
- Dashboard drill-down
- Alerts

Makes the app useful on the line every day.

Wave 3

Planning

- Production lines
- Plant calendar
- Line load chart
- Shop-floor scanning

From tracking orders to planning capacity.

Wave 4

Integration

- Company sign-in
- Bill of materials
- ERP and customer orders
- Lot traceability
- Schedule board, material requirements

Connects the plant to the rest of the business.

Each function would become its own work item — brief, approved plan, basic, database and detailed design, implementation and tests — like Screens A–C.

Sizes are rough guesses from the current codebase, before any requirements work. Source: [SateraitoOfficeVN/ProductionManagementAI](#).